

# Multi-Objective Selection for Auditable Redistricting Plans

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## Abstract

This draft frames selection among objective trade-offs after construction, ensemble sampling, SMC, or Pareto search. It emphasizes how selected plans should carry objective vectors, selection rationale, and RPLAN/RCTX audit lineage without overclaiming neutrality. A selected plan is treated as a decision record: candidate set, objective definition, selection rule, and final audit package must travel together.

## 1 Introduction

U.14 is a methodology paper about choosing among trade-offs. It treats multi-objective outputs as decision surfaces that require explicit selection rationale and audit lineage.

The paper extends U.12 and U.13 into the selection setting. A Pareto frontier, ensemble cloud, or SMC sample can reveal alternatives; it does not choose among them without a rule. The selection rule may be technical, policy-driven, or administrative, but it must be recorded because it changes the claim attached to the exported plan.

## 2 Selection Framework

Selection should record objective definitions, weights or dominance rules, frontier position, candidate set provenance, and the reason a final plan was chosen. The goal is transparency, not automatic neutrality.

The selected plan should therefore be accompanied by the rejected context. When only the final assignment is exported, readers cannot tell whether the plan was representative of the frontier, an extreme point, a compromise under declared weights, or a policy choice after inspection.

## 3 Audit Artifacts

Selected plans should export objective vectors, candidate lineage, selection parameters, and final assignments through the RPLAN/RCTX fixed point. Audit checks verify declared artifacts, not the normative correctness of the chosen trade-off.

| Selection input  | Required record                                 | Main risk if omitted      |
|------------------|-------------------------------------------------|---------------------------|
| Objective vector | Metric definitions and computation command      | Unclear trade-off meaning |
| Candidate set    | Generator, seed, filters, and count             | Cherry-picking ambiguity  |
| Dominance rule   | Pareto, weighted score, threshold, or tie-break | Hidden policy choice      |
| Chosen plan      | Selection rationale and rank/frontier position  | False neutrality claim    |
| Export package   | RPLAN/RCTX plus objective sidecars              | Unreproducible final plan |

Table 1: Minimum record for auditable multi-objective selection.

The audit package should include enough metadata to replay the decision at the artifact level: candidate identifiers, objective values, dominance or scoring rule, tie breaks, and the final selected RPLAN. This creates a reviewable record even when the selection rule itself is contested.

## 4 Current Implementation Boundary

Multi-objective selection now has a concrete selected-frontier package pattern through `bisect-pareto`: selected index, objective vector, RPLAN/RCTX sidecars, manifest metadata, and audit-certificate path. The selection record must travel with the plan because the same assignment can carry different meaning under a different candidate set or objective definition.

This paper should not imply that a selected plan is neutral because it came from a frontier. The defensible claim is narrower: the package records which candidate was selected, from what declared set, under what objective vector and selection rule, and whether the selected plan passed the declared verifier.

## 5 Limitations

Multi-objective selection can expose trade-offs but cannot decide legal or political values by itself. Empirical claims require recorded candidate sets, commands, and objective computations.

The current draft is methodological. Publication readiness requires concrete examples from exported-plan, ensemble, SMC, or Pareto workflows, including the commands that produced candidate sets and the sidecars that preserved objective vectors. Until then, claims should remain about audit discipline rather than measured selection performance.

## 6 Conclusion

U.14 turns Pareto, SMC, ensemble, and optimization outputs into auditable selection decisions by preserving both the candidate set and the selection rationale.

This makes multi-objective work legible: the portfolio can expose trade-offs, choose a final plan, and still show the reader what choice was made and what evidence traveled with it.

## References

BISECT Project. U.14 multi-objective selection plan, 2026. BISECT internal paper plan.